Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	*(a)*			**Any two × (1) from the following:** • Cables can be thinner [because there are two paths for the current] • Each part of the cable carries less current [because the current flows two ways] • Sockets can be {placed / added} anywhere on the ring • Same voltage [accept 230 V or 240 V] for all sockets **or** separate switching	2			2		
	(b)	(i)		To carry the {current / power / energy} {to the appliance / to the socket / at high voltage} NB not: ‘electricity’	1			1		
		(ii)		Carries current [safely] to ground **or** prevents {[electric] shock / electrocution} (1) if the metal of an appliance frame becomes <u>live</u> [or <u>live</u> wire touches the metal case of an appliance] (1)	2			2		
	(c)	(i)		a.c. is a current that [continuously] changes <u>direction</u>(1) a.c. is a current that [continuously] changes in <u>magnitude</u> [or size](1) (Accept converse answers about d.c.)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Substitution: in $I = \frac{P}{V}$; current = $\frac{2\,760}{230}$ (1) = 12 [A] (1) [12 by itself on answer line $\rightarrow$ 0 marks; with above working $\rightarrow$ 2 marks] Substitution: in $R = \frac{V}{I}$; resistance = $\frac{230}{12\ \textbf{ecf}}$ (1) = 19.17 [Ω] (1) [accept 19.2 Ω or 19 Ω, not 19.6 Ω] ***OR for third and fourth marks:*** $P=I^2R$ so substitution: 2 760 = $R \times 12^2$ (1) = 19.17 [Ω] (1) **or** for all the marks Use of $P = \frac{V^2}{R}$ (1) Substitution: $2760 = \frac{230^2}{R}$ (1) Manipulation: $R = \frac{230^2}{2760}$ (1) Answer = 19.17 Ω (1)	1 1	1 1		4	4	
				Question 2 total	**9**	**2**	**0**	**11**	**4**	**0**